

Plan4ARI

Review of Activity A2

Low-code programming with generative and deliberative AI

Plan4ARI

Methodological update and technical specification alignment

Draft for technical discussion / 23 September 2026



Table: Mesi-persona per Attività 2.

WP	Soggetto	Fund. Research	Ind. Research	Exp. Development	Totale
WP2	HI	3.81	62.40	0.00	66.21
	AI	1.96	3.91	0.00	5.87
	RO	11.05	23.38	0.00	34.43
	<i>Totale WP</i>	<i>16.81</i>	<i>89.70</i>	<i>0.00</i>	<i>106.51</i>

Table: Mesi-persona per WP - anno 1.

WP	Soggetto	Fund. Research	Ind. Research	Exp. Development	Totale
WP2	HI	1.90	20.84	0.00	22.75
	AI	0.98	0.78	0.00	1.76
	RO	5.52	7.01	0.00	12.54
	<i>Totale WP</i>	<i>8.40</i>	<i>28.64</i>	<i>0.00</i>	<i>37.04</i>

Application objective

Enable process experts to program robotic applications through natural language.

Integrate the modules into COMAU MI.RA/Dexter.

Activity A2: **M1–M36**.

Targets

- **66%** reduction in programming time.
- Programming by workers without prior programming training.
- A fleet of **3 mobile manipulators** or **4 industrial robotic arms**.

Source: Annex A, Section A.3.4, p. 10, and Activity 2, p. 36. The baseline uses “not pre-trained worker”.

Original approach and proposed revision

Stage	Original approach	D2.1 v0.11 proposal
Knowledge	Knowledge base from process descriptions	Capability registry with evidence and verification
Formalisation	PDDL generation using a lightweight LLM	Typed intermediate representation and compilation
Planning	BTs and digital-twin optimisation	Solve and validate the plan before BT generation
Optimisation	MCTS search in simulation	Solver portfolio, scheduling and geometric checks
Execution	BT execution and online replanning	Skill grounding, effect verification and recovery

Source: Annex A, Section A.3 and Tasks 2.2–2.5; D2.1 v0.11, Sections 1.1, 6–8. Changes are proposed.

Symbolic plans must correspond to executable capabilities

- An available ROS2 interface does not, by itself, specify a skill's preconditions, effects or constraints.
- The capability registry combines runtime interfaces, documentation, MI.RA/Dexter blocks and verification evidence.
- Grounding binds each planned action to a concrete skill with consistent parameters and resources.
- Runtime monitoring compares expected effects with observed state and triggers recovery when required.

Source: D2.1 v0.11, Sections 1 and 7; Research Proposal, Sections 2.1–2.2. Summary of the proposed research.

Three research directions in the A2 proposal

Direction	Research question	Proposed task mapping
1. Available domain	How can a request become a valid plan grounded in ROS2 skills?	T2.2, T2.3 and T2.5
2. Skill discovery	How can a domain be built or adapted from the system and its evidence sources?	T2.2, supported by T2.6
3. Scene graphs	How can relations between parts support disassembly planning and state verification?	T2.3–T2.5, with A3/A4 interfaces

Direct BT generation from scene graphs is an exploratory research direction. Its integration into the validated planning pipeline requires explicit definition.

Source: Research Proposal, Sections 2.1–2.3; D2.1 v0.11, Sections 7 and 10. Task mapping proposed for this review.

Proposed functional architecture

Stage	Output	Required checks
Operational knowledge	Skill registry and scene state	Provenance, availability and consistency
Interpretation	Typed goals and constraints	Ambiguity, objects and types
Formalisation	PDDL domain and problem	Model syntax and semantics
Planning	Plan and resource allocation	Constraints, validity and solution quality
Execution preparation	BT and executor bindings	Parameters, resources and twin feasibility
Execution	Execution trace and updated state	Preconditions, effects and timing

Source: D2.1 v0.11, Sections 7–8. Proposed functional sequence, not an implemented architecture.

Proposed revision of T2.2–T2.4

Task and period	Proposed change	Required evidence
T2.2 M13–M24	Capability discovery, lightweight models, typed representation and PDDL compilation	Verified registry, reference problems and comparison with direct generation
T2.3 M18–M24	BTs derived from validated plans. Explicit temporal handling using STN/STNU	Plan-to-BT correspondence and variable-duration tests
T2.4 M22–M30	Solver selection by problem type. Twin-based feasibility and cost estimation	Quality against a baseline, motion feasibility and computational cost

Source: Annex A, p. 37; D2.1 v0.11, Sections 8 and 10. Baseline task periods retained in this proposal.

Proposed revision of T2.5–T2.8

Task and period	Proposed change	Required evidence
T2.5 M28–M36	State and effect checks, bounded plan repair and replanning	Perturbation tests, recovery times and required interventions
T2.6 M28–M36	MI.RA/Dexter block binding and module integration	Integrated workflows and interface tests
T2.7 M13–M36	Versioning of software, models, prompts and benchmarks	Reproducible tests and release traceability
T2.8 M7–M12	Initial benchmark definition, with proposed ongoing evaluation	Protocol, test cases and end-user evaluation

Source: Annex A, pp. 37–38; D2.1 v0.11, Section 10. Extending T2.8 requires explicit schedule alignment.

Deliverables and schedule dependencies

Deadline	Baseline commitment	Issue to resolve
D2.1 / M12	Scientific plan and first running mockup of the low-code module	Confirmed for M12. D2.1 v0.11 includes the mockup described in the Research Proposal and its demonstration evidence
D2.2 / M24	First time-variance-aware planner running in MI.RA/Dexter	Minimum integration is needed by M24. T2.6 starts at M28; the v0.9 roadmap places integration at M25–M30
D2.3 / M36	Optimal task and motion planning with online replanning	Link the demonstrator to KPI evidence and acceptance criteria

Source: Annex A, pp. 38 and 46; D2.1 v0.11, “Purpose and boundary” and Section 10. Issues identified through comparison.

D2.1 includes the scientific plan and the running mockup

- Operator GUI, natural-language request, optional context and an available PDDL domain.
- JSON problem generation, deterministic PDDL compilation, validation and repair.
- Planner portfolio and action grounding through ROS2 introspection and PlanSys2 executors.
- Initial single-robot pick-and-place demonstration with versioned artefacts and execution evidence.

Skill-registry generation and mechanical scene graphs extend the research programme. The M12 report will state which prototype functions are implemented and tested.

Source: Research Proposal, Section 2.1, Figure 1 and Section 3; D2.1 v0.11, Section 6.11. M12 availability confirmed in the project update.

Submission target: 23 October 2026

- Paper on the mockup, neuro-symbolic architecture and initial experimental evaluation.
- Candidate journal: IEEE Transactions on Automation Science and Engineering (IEEE T-ASE) or IEEE Transactions on Industrial Informatics (IEEE TII).
- Complete experiments, reproducibility materials, failure analysis and co-author review before submission to one selected journal.
- Record submission acknowledgement separately from any subsequent acceptance or publication.

The calendar target is within one month of the 23 September update. It does not redefine the project M12 date.

Source: Project planning update, 23 September 2026; D2.1 v0.11, Sections 8.7 and 10.5. Planned submission.

Document corrections for discussion

Issue	Evidence	Proposed correction
T2.1 period	Original specification: M7–M12. Revised A2 table: M1–M12	Aligned to M1–M12 in D2.1 v0.11, following the proposed specification revision
T2.4 input	Text assigns BT output to T2.2. BT generation belongs to T2.3	Correct the reference and specify the actual input
M24/M36 milestones	The milestone list still cites D2.1	Check references to D2.2 and D2.3. Also clarify the M48 reference
T2.8 period	M7–M12, although the text includes platform evaluation	Separate initial benchmark design from later evaluation campaigns

Source: Annex A, pp. 37–38 and 46; technical specification, Activity 2 and milestones; D2.1 v0.11, Sections 2.3 and 10.

Baseline commitments

- Programming time reduction: 66%.
- Workers without prior training.
- 3 mobile manipulators or 4 arms.
- Knowledge base: 10 contexts.
- PDDL, BT and task planning: “+100 contexts” for each category.

New thresholds in v0.9 are proposals. The definition of a “context” and the counting protocol require agreement.

Source: Annex A, Section A.3.4, p. 10; D2.1 v0.11, Section 9 and metric table.

Additional D2.1 metrics

- Capability discovery accuracy.
- Model and plan validity.
- Actions grounded in executable skills.
- Planning quality and latency.
- Robustness and human interventions.

Measure the full process up to an executable workflow

- 1 Define cases, initial conditions and correct outcomes with domain experts.
- 2 Compare direct generation, a validator-based pipeline and a capability-registry-based pipeline.
- 3 Measure initial cell configuration, task programming and human correction effort separately.
- 4 Test duration changes, missing objects and unavailable skills.
- 5 Evaluate the integrated workflow with operators at the target robot fleet scale.

Programming time reduction: $100 (T_{\text{baseline}} - T_{\text{A2}}) / T_{\text{baseline}}$.

Use identical tasks and completion criteria in both conditions.

Source: Review proposal based on Annex A, Section A.3.4 and T2.8; D2.1 v0.11, Section 9; Research Proposal, Section 3.

Interfaces and responsibilities to agree

Interface	Required agreement
CNR / planning specialists	Capability models, formalisation, solvers, validation and prototypes
COMAU / MI.RA/Dexter	Block catalogue, interfaces, industrial requirements and integration deadlines
A3 / A4	Perceived state, motion feasibility checks and execution feedback
A6 / users	Comprehensibility, workload, interventions and task completion
A2 lead / PI	Methodological choices, ownership, acceptance criteria and change management

Proposed boundary: cognitive functions in the Open IPC; hard-real-time control and safety-critical functions in the Core IPC.

Source: D2.1 v0.11, Sections 2.4 and 10, including the roles table. Proposed assignments require confirmation.

Potential COMAU design contributions (1/3)

Evidence status: contributions to confirm and document, not verified completed activities.

Potential COMAU contribution	Effect on the method and mockup	Evidence to retrieve
Dexter programming workflow analysis	Identify decisions that currently require specialist expertise and those suitable for automation.	Annotated demonstrations, example workflows, meeting notes and emails.
Block-library analysis	Specify how a block becomes a plannable skill: parameters, preconditions, effects, resources and errors.	Module descriptions, configuration examples and technical annotations.
Industrial requirements definition	Inform structured generation, validation and explicit handling of missing capabilities.	Requirements, architecture review comments and documented decisions.

Source: Contribution framework based on the Host Institution role and the project team's description of Dexter. D2.1, Section 9.

Potential COMAU design contributions (2/3)

Evidence status: contributions to confirm and document, not verified completed activities.

Potential COMAU contribution	Effect on the method and mockup	Evidence to retrieve
Use-case definition	Define realistic process sequences, constraints, variants and mockup failure conditions.	Use-case specification, process examples and test scenarios.
Future interface design	Specify the mapping between symbolic actions, ROS2 skills and Dexter modules for subsequent integration.	Interface diagrams, preliminary mappings and compatibility assessments.
Mockup review and acceptance criteria	Assess industrial relevance and define how the usefulness of the prototype will be measured.	Demo feedback, annotated tests, issue lists and resulting corrections.

Source: Contribution framework based on the Host Institution role and the project team's description of Dexter. D2.1, Section 9.

In other words

COMAU should

- Define the Knowledge base and skill registry for the low-code programming framework.
- Define the operational capability models and their verification evidence.
- Define the use cases for the low-code programming framework.
- Design the interface for the framework.
- Review and define the acceptance criteria for the mockup.

<https://gist.github.com/karpathy/442a6bf555914893e9891c11519de94f>

Working text for the revised technical specification

Activity A2 will develop a low-code programming framework combining generative AI and deliberative planning, intended for process experts and integrated into MI.RA/Dexter.

The framework will translate natural-language requests into verifiable planning models, using an operational capability registry and a scene-state representation. Planned actions will be bound to executable skills and checked against temporal constraints, resource availability and motion feasibility.

Execution will include effect monitoring and replanning. Evaluation will document the baseline targets for programming time reduction, operator accessibility and multi-robot scale.

Source: Proposed wording based on the technical specification, Activity 2, and D2.1 v0.11. This is not approved text.

Decisions required for the revision

- ① Confirm the pipeline based on a capability registry, validated models and executable skill bindings.
- ② Define the role of exploratory work: disassembly scene graphs, MCTS and adapter generation.
- ③ Confirm the M12 mockup evidence package and the manuscript submission plan.
- ④ Align minimum integration with D2.2/M24 and clarify continued T2.8 evaluation.
- ⑤ Agree on benchmarks, ownership and acceptance criteria, keeping baseline KPIs distinct from additional metrics.

Source: Summary of open issues identified through the document comparison.

Sources and document status

- **Annex A, FISA-2024-00099**: technical reference for objectives, KPIs, tasks and deliverables. Main passages: p. 10, pp. 36–38 and p. 46.
- **Plan4ARI v3, final technical specification**: editable Activity 2 text and deliverable list used for comparison.
- **D2.1 Updated Scientific Plan, v0.9**: revised internal draft including the M12 mockup, methodological updates and manuscript submission plan.
- **A2 Research Project Proposal, 9 July 2026**: research proposal covering neuro-symbolic planning, skill discovery and scene graphs.
- **budget_COMAU_v6.xlsx, Budget Per Year**: annual partner costs and funding; hypothetical effort at EUR 6,100/PM for COMAU and EUR 5,200/PM for CNR.
- **Annexes B, C and D**: financial framework, project terms and identifiers. No budget amendment is proposed.

This presentation compares the supplied documents. It does not include a new external verification of the literature cited in D2.1.